STAT8561 - Linear Statistical Analysis I

2026 Fall Lecture Notes

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Preface

The topic of this course includes statistical inference, Multivariate normal distribution, distribution of quadratic forms, linear models, regression models and experimental design models.

Prerequisites

Math 4751/6751 Mathematical Statistics I and Math 4752/6752 Mathematics and Statistics II.

Instructor

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Office Hour

9:00 - 12:00 on Wednesday or by appointment

Grade Distribution

- Homework – 40%
- Exam 1 – 20%
- Exam 2 – 20%
- Final – 20%
- Attendance – *7%

Assignment

- ☐ A1, TBA

Exam

- TBA

Topics and Corresponding Lectures

Those chapters are based on the lecture notes. This part will be updated frequently.

Status	Chapter	Topic	Lecture
		Welcome and Overview	1
	Ch. 1	Math Notation	2–3
	Ch. 2	Least Squares Estimation	3–

Recommended Textbooks

- Ding, Peng. *Linear Model and Extensions.* CRC Press, 2027. arXiv link
- Montgomery, Douglas C., Elizabeth A. Peck, and G. Geoffrey Vining. *Introduction to Linear Regression Analysis.* John Wiley & Sons, 2021. (Montgomery et al. 2021)
- Seber, George AF, and Alan J. Lee. *Linear Regression Analysis.* John Wiley & Sons, 2003. (Seber and Lee 2003)

Side Readings

- Montgomery, Douglas C. (2017). *Design and Analysis of Experiments.* John Wiley & Sons. (Montgomery 2017)

Introduction

This is an introduction to the course for linear statistical analysis. It will give you an overview of what we will be covering in the course and how to get the most out of it. We will be covering a wide range of topics in linear statistical analysis,

- simple linear regression
- multiple linear regression
- generalized linear models
- mixed effects models
- quantile regression.

We will also be discussing the assumptions underlying these models and how to check them.

The course will be structured around lectures, homework assignments, a midterm and a final.

To get the most out of this course, it is important to attend all lectures and complete all homework assignments. It is also important to ask questions and participate in class discussions. The more you engage with the material, the more you will learn.

I am looking forward to a great semester and I hope you are too!

To illustrate the concepts we will be covering in this course, let's consider a simple example. Suppose we have a data set of heights and weights of individuals. We want to understand the relationship between height and weight, and we can use linear regression to model this relationship.

```
# Load necessary libraries
library(ggplot2)
library(dplyr)
# Create a sample data set

set.seed(8561)

theme_set(theme_minimal())

n <- 100
heights <- rnorm(n, mean = 170, sd = 10)
weights <- 0.5 * heights + rnorm(n, mean = 0, sd
```

```r
 = 5)
df <- data.frame(heights, weights)
head(df, 5)
```

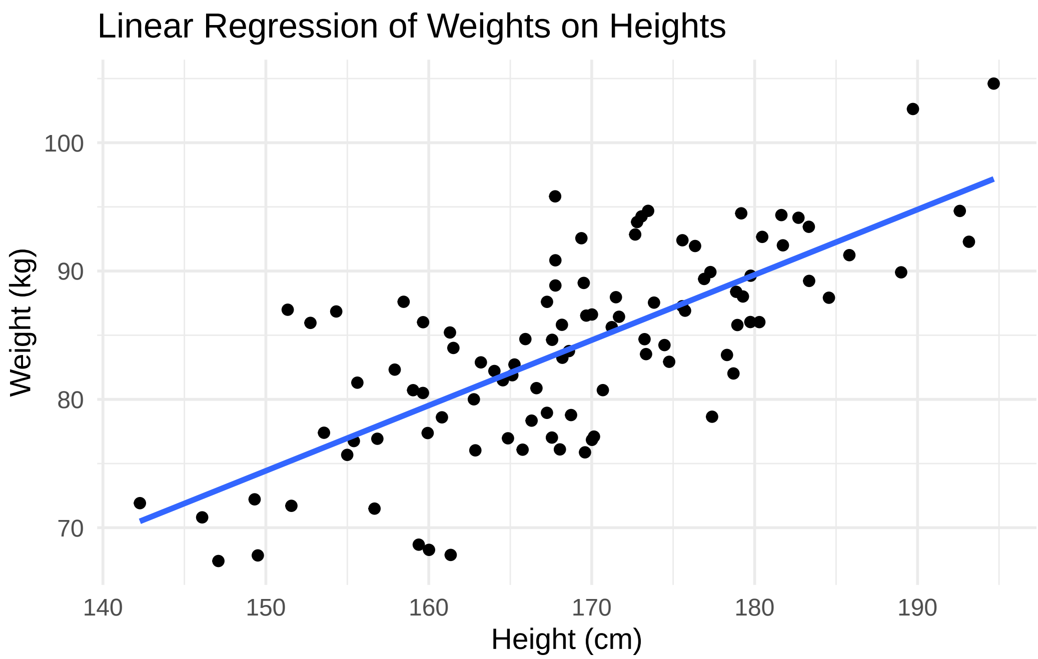
```
   heights  weights
1 179.2812 88.01599
2 168.0474 76.10017
3 178.3029 83.46016
4 178.7006 82.01763
5 172.6678 92.84592
```

```r
# Fit a linear regression model
model <- lm(weights ~ heights, data = df)
# Summarize the model
summary(model)
```

```
Call:
lm(formula = weights ~ heights, data = df)

Residuals:
     Min       1Q   Median       3Q      Max 
-12.3303  -3.8998  -0.2191   3.2380  12.3478 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -1.92852    8.22226  -0.235    0.815    
heights      0.50907    0.04857  10.481   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.251 on 98 degrees of freedom
Multiple R-squared:  0.5285,	Adjusted R-squared:  0.5237 
F-statistic: 109.9 on 1 and 98 DF,  p-value: < 2.2e-16
```

```r
# Plot the data and the fitted line
ggplot(df, aes(x = heights, y = weights)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Linear Regression of Weights on Heights",
       x = "Height (cm)",
       y = "Weight (kg)")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



In this example, we generated a data set of heights and weights, fitted a linear regression model to the data, and visualized the relationship between height and weight. This is just a simple example, but it illustrates the types of analyses we will be doing in this course. We will be covering much more complex models and data sets as we progress through the semester.

Throughout the course, we will be using **R** programming software.

1 Matrix Representation and Notation

In this chapter, we introduce the mathematical language and statistical framework that will be used throughout the course. Our focus is on the notation of vectors and matrices, vectors of random variables, expectation and covariance operators, and the basic form of the linear regression model.

> **Learning Objectives**
>
> By the end of this chapter, students should be able to:
>
> - use standard matrix notation for linear statistical models;
> - distinguish between scalars, vectors, matrices, random variables, and random vectors;
> - compute expectations and covariance matrices for random vectors;
> - interpret the linear regression model in matrix form;
> - understand why projection and least squares will play a central role in this course.

Reading

Recommended reading for this chapter:

- Seber and Lee (2003), Chapter 1:
  - 1.1 Notation
  - 1.2 Statistical Models
  - 1.3 Linear Regression Models
  - 1.4 Expectation and Covariance Operators
- Optional preview:
  - Chapter 2: Multivariate Normal Distribution

1.1 Why Linear Statistical Analysis?

Linear statistical analysis is one of the central foundations of graduate statistics. Many methods that at first look different are built on the same underlying structure:

- regression
- analysis of variance
- analysis of covariance
- prediction
- model comparison
- and parts of generalized linear modelling

A major goal of this course is to see these topics under a unified framework.

1.2 A unifying point of view

A large part of the course can be summarized by the model

$$Y = X\beta + \varepsilon,$$

where

- Y is a response vector,
- X is a design matrix,
- β is an unknown parameter vector,
- ε is a random error vector.

This compact expression contains a great deal of statistical structure. Over the semester, we will study how to estimate β, quantify uncertainty, test hypotheses, diagnose model failures, and make predictions.

1.3 Basic Notation

Scalars, vectors, and matrices

We use the following conventions throughout the course:

- scalars are written in lowercase italic letters, such as a, b, n;
- vectors are written in bold lowercase letters, such as $\mathbf{x}$, $\mathbf{y}$;
- matrices are written in bold uppercase letters, such as $\mathbf{X}$, $\mathbf{A}$;
- random variables are often written in uppercase letters, such as Y;
- realizations of random variables are written in lowercase letters, such as y.

Univariate case

A random variable is **univariate** if it takes values in one dimension. We write

$$X \in \mathbb{R}.$$

A realization, or observed value, of X is denoted by x.

Some common quantities associated with a univariate random variable X include:

- **Mean**
$$\mu = \mathbb{E}(X).$$
- **Variance**
$$\sigma^2 = \mathrm{Var}(X) = \mathbb{E}\left[(X-\mu)^2\right].$$
- **kth moment**
$$\mathbb{E}(X^k).$$
- **Cumulative distribution function (CDF)**
$$F(x) = P(X \leq x).$$
- **Probability density function (PDF)**, for a continuous random variable,
$$f(x).$$
- **Probability mass function (PMF)**, for a discrete random variable,
$$p(x) = P(X = x).$$
- **Quantile**
$$q_\alpha = F^{-1}(\alpha),$$
where $0 < \alpha < 1$.

For a random sample of size n, we write

$$X_1, X_2, \dots, X_n,$$

with observed values

$$x_1, x_2, \dots, x_n.$$

The sample mean is

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i,$$

and the sample variance is

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

Multivariate case

A column vector in $\mathbb{R}^n$ is written as

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = (x_1, x_2, \dots, x_n).$$

An $n \times p$ matrix is written as

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}.$$

Transpose, inverse, and rank

If $\mathbf{A}$ is a matrix, then:

- $\mathbf{A}^\top$ denotes its transpose;
- $\mathbf{A}^{-1}$ denotes its inverse, when it exists;
- $\operatorname{rank}(\mathbf{A})$ denotes its rank;
- $\mathbf{I}_n$ denotes the $n \times n$ identity matrix.

1.3.1 Inner product and norm

For vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the inner product is

$$\mathbf{x}^\top \mathbf{y} = \sum_{i=1}^{n} x_i y_i.$$

For vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the inner product is

$$\|\mathbf{x}\| = \sqrt{\mathbf{x}^\top \mathbf{x}}.$$

These ideas are fundamental because least squares estimation is based on minimizing squared Euclidean distance.

A random vector is a vector whose entries are random variables. For example,

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}$$

is an n-dimensional random vector.

In linear models, the response is naturally treated as a random vector.

The mean vector of $\mathbf{Y}$ is

$$\mathbb{E}[\mathbf{Y}] = \begin{pmatrix} \mathbb{E}[Y_1] \\ \mathbb{E}[Y_2] \\ \vdots \\ \mathbb{E}[Y_n] \end{pmatrix}.$$

We often write $\mathbb{E}[\mathbf{Y}] = \mu$.

The covariance matrix of $\mathbf{Y}$ is

$$\mathrm{Var}(\mathbf{Y}) = \mathbb{E}\left[(\mathbf{Y} - \mu)(\mathbf{Y} - \mu)^\top\right].$$

Equivalently,

$$\mathrm{Var}(\mathbf{Y}) = \mathbb{E}[\mathbf{Y}\mathbf{Y}^\top] - \mu\mu^\top.$$

Suppose $\mathbf{Y} = (Y_1, Y_2, Y_3)$.

Then the corresponding covariance matrix is

$$\mathrm{Var}(\mathbf{Y}) = \begin{pmatrix} \mathrm{Var}(Y_1) & \mathrm{Cov}(Y_1, Y_2) & \mathrm{Cov}(Y_1, Y_3) \\ \mathrm{Cov}(Y_2, Y_1) & \mathrm{Var}(Y_2) & \mathrm{Cov}(Y_2, Y_3) \\ \mathrm{Cov}(Y_3, Y_1) & \mathrm{Cov}(Y_3, Y_2) & \mathrm{Var}(Y_3) \end{pmatrix}.$$

1.3.2 Expectation and Covariance Operators

If $\mathbf{A}$ is a constant matrix and $\mathbf{b}$ is a constant vector, then

$$\mathbb{E}[\mathbf{AY} + \mathbf{b}] = \mathbf{A}\mathbb{E}[\mathbf{Y}] + \mathbf{b}.$$

This is one of the most useful identities in the course.

If $\mathbf{A}$ is a constant matrix, then

$$\mathrm{Var}(\mathbf{AY}) = \mathbf{A}\,\mathrm{Var}(\mathbf{Y})\,\mathbf{A}^\top.$$

More generally, if $\mathbf{Y}$ and $\mathbf{Z}$ are random vectors, then

$$\mathrm{Cov}(\mathbf{AY}, \mathbf{BZ}) = \mathbf{A}\,\mathrm{Cov}(\mathbf{Y}, \mathbf{Z})\,\mathbf{B}^\top.$$

These formulas are essential for deriving the variance of estimators later.

If $Y_1, \ldots, Y_n$ are independent and each has variance σ^2, then

$$\mathrm{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n.$$

This is the most common starting assumption in classical linear regression.

1.4 Statistical Models

A statistical model is a set of probability distributions that may plausibly describe the data-generating mechanism.

1.4.1 General idea

Suppose we observe data y from a random quantity Y. A model introduces assumptions about the distribution of Y, often indexed by an unknown parameter θ.

For example:

$$Y \sim N(\mu, \sigma^2)$$

with unknown parameters μ and σ^2.

In regression, the model is not only about the marginal distribution of the response but also about how the mean changes with explanatory variables.

1.4.2 Deterministic part and random part

A useful way to think about a statistical model is:

$$\text{data} = \text{systematic part} + \text{random part}.$$

For linear regression, this becomes

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

Here,

- $\beta_0 + \beta_1 x_i$ is the systematic part;
- ε_i is the random part.

1.5 The Linear Regression Model

Perhaps the most simple model is the *simple linear regression* where there are only **one** independent variable and **one** response variable.

A simple linear regression model is defined as

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \dots, n,$$

with the typical assumption as

$$\mathbb{E}[\varepsilon_i] = 0, \qquad \mathrm{Var}(\varepsilon_i) = \sigma^2, \qquad \mathrm{Cov}(\varepsilon_i, \varepsilon_j) = 0 \text{ for } i \neq j.$$

Matrix form

We can write the model compactly as

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon,$$

where

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \qquad \mathbf{X} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}, \qquad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}, \qquad \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

This notation will allow us to treat simple regression, multiple regression, ANOVA, and ANCOVA in one common language.

Under the same assumption,

$$\mathbb{E}[\varepsilon] = \mathbf{0} \qquad \text{and} \qquad \mathrm{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n,$$

we have the following.

$$\mathbb{E}[\mathbf{Y}] = \mathbf{X}\beta, \quad \mathrm{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n.$$

These are immediate consequences of the expectation and covariance rules above.

1.5.1 Geometry Preview

A central idea in linear regression is projection.

The fitted values $\hat{\mathbf{Y}}$ will later be obtained by projecting $\mathbf{Y}$ onto the column space of $\mathbf{X}$.

The column space of $\mathbf{X}$ is

$$\mathcal{C}(\mathbf{X}) = \{\mathbf{X}\beta : \beta \in \mathbb{R}^p\}.$$

This is the set of all mean vectors that the model can represent.

1.6 Why projection matters

The least squares estimator chooses $\hat{\beta}$ so that

$$\|\mathbf{Y} - \mathbf{X}\beta\|^2$$

is minimized.

So regression is not only algebra. It is also geometry.

Suppose we observe the following data:

x_i	0	1	2	3
y_i	1	3	3	5

Then

$$\mathbf{Y} = \begin{pmatrix} 1 \\ 3 \\ 3 \\ 5 \end{pmatrix}, \qquad \mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}.$$

We will learn next chapter that the least squares estimator is

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

For now, the main point is to understand how the model is written and how the data are represented in matrix form.

R code

```
x <- c(0, 1, 2, 3)
y <- c(1, 3, 3, 5)

mod <- lm(y ~ x)
summary(mod)
```

```
Call:
lm(formula = y ~ x)

Residuals:
   1    2    3    4 
-0.2  0.6 -0.6  0.2 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)  
(Intercept)   1.2000     0.5292   2.268   0.1515  
x             1.2000     0.2828   4.243   0.0513 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6325 on 2 degrees of freedom
Multiple R-squared:    0.9,	Adjusted R-squared:   0.85 
F-statistic:    18 on 1 and 2 DF,  p-value: 0.05132
```

```r
# Inspecting the design matrix
model.matrix(mod)
```

```
  (Intercept) x
1           1 0
2           1 1
3           1 2
4           1 3
attr(,"assign")
[1] 0 1
```

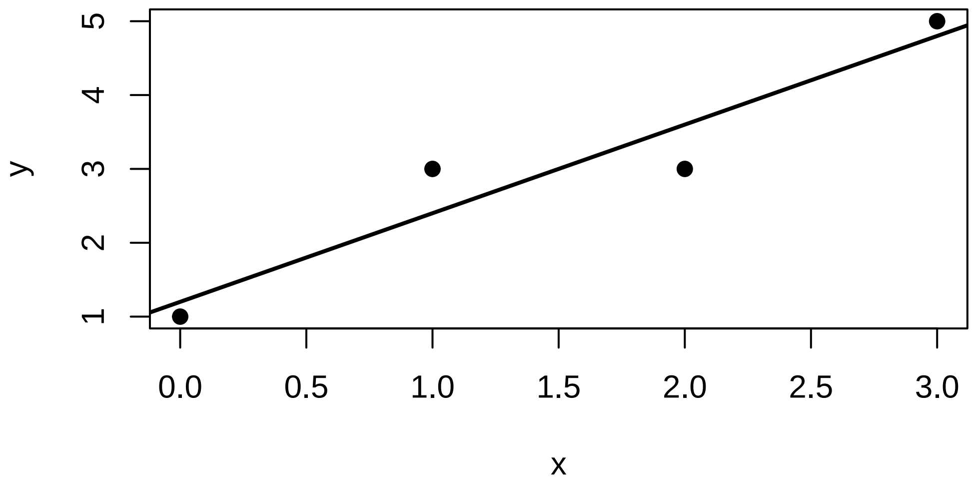
```r
# Fitted values and residuals
fitted(mod)
```

```
  1   2   3   4 
1.2 2.4 3.6 4.8 
```

```r
residuals(mod)
```

```
   1    2    3    4 
-0.2  0.6 -0.6  0.2 
```

```r
# quick plot
plot(x, y, pch = 19, xlab = "x", ylab = "y")
abline(mod, lwd = 2)
```



1.7 Discussion

1. Why is it helpful to write regression models in matrix form rather than only scalar notation?
2. What does the covariance matrix tell us that separate variances do not?
3. What is the interpretation of the column space of $\mathbf{X}$?
4. In what sense is regression a projection problem?

1.8 Practice Problems

Conceptual

1. Explain the difference between a random variable and a random vector.
2. Explain why $\mathrm{Var}(\mathbf{Y})$ must be a symmetric matrix.

3. Give an example of a statistical model outside regression.

Computational

Let

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix}$$

with

$$\mathbb{E}[\mathbf{Y}] = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad \mathrm{Var}(\mathbf{Y}) = \begin{pmatrix} 4 & 1 \\ 1 & 9 \end{pmatrix}.$$

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}.$$

Compute:

1. $\mathbb{E}[\mathbf{AY} + \mathbf{b}]$
2. $\mathrm{Var}(\mathbf{AY})$

Regression setup

For the model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,$$

write out the matrices $\mathbf{Y}$, $\mathbf{X}$, β, and ε for $n = 5$ observations.

1.9 Take home Practice

Complete the following:

- review matrix multiplication and transpose rules;
- derive $\mathbb{E}[\mathbf{AY} + \mathbf{b}]$ from first principles;
- derive $\mathrm{Var}(\mathbf{AY})$ using the definition of covariance;
- write the simple linear regression model in matrix form for a dataset of your choice;
- fit a simple regression in R and report:
- the estimated coefficients,

- fitted values,
- residuals,
- and a scatterplot with the fitted line.

1.10 Summary

This chapter introduced the notation and basic probabilistic tools needed for the rest of the course. We defined random vectors, mean vectors, covariance matrices, and the matrix form of the linear regression model. These ideas will support everything that follows.

Next chapter, we will study least squares estimation and the geometry of projection in more detail.

For some optional review of the matrix algebra, see Matrix Algebra Review.

2 Least Squares Estimation

In this chapter, we study one of the fundamental ideas in linear models: **least squares estimation**.

Recall that the linear model is

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon.$$

The unknown parameter vector β determines the mean structure of the response. Given observed data, our first goal is to estimate β.

The central idea of least squares is simple:

> Choose the value of β for which the fitted response $\mathbf{X}\beta$ is as close as possible to the observed response $\mathbf{Y}$.

This optimization problem leads to the **normal equations**, the **least squares estimator**, and a useful geometric interpretation in terms of **orthogonal projection**.

> Learning Objectives
>
> By the end of this week, students should be able to:
>
> - formulate least squares estimation as an optimization problem;
> - define the residual sum of squares;
> - derive the normal equations;
> - obtain the least squares estimator when $\mathbf{X}$ has full column rank;
> - explain the role of the column space $\mathcal{C}(\mathbf{X})$;
> - interpret fitted values as an orthogonal projection;
> - explain why residuals are orthogonal to the model space;
> - define and interpret the hat matrix and residual-maker matrix;
> - verify symmetry and idempotence of projection matrices;
> - explain the decomposition $\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}$;
> - distinguish the uniqueness of $\hat{\beta}$ from the uniqueness of the fitted values.

Reading

Recommended reading for this week:

- Seber and Lee:
  - Chapter 3: sections on least squares estimation
- Montgomery, Peck, and Vining:
  - introductory sections on estimation in linear regression

Recap of the Linear Model

Recall from Week 1, the linear model can be written as

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon,$$

where

- $\mathbf{Y}$ is an $n \times 1$ response vector;
- $\mathbf{X}$ is an $n \times p$ design matrix;
- β is a $p \times 1$ parameter vector;
- ε is an $n \times 1$ error vector.

The design matrix may be written as

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix},$$

where $\mathbf{x}_i \in \mathbb{R}^p$ contains the predictors associated with observation i.
Thus,

$$Y_i = \mathbf{x}_i^\top \beta + \varepsilon_i, \qquad i = 1, \ldots, n.$$

Under the classical linear model, we often assume

$$\mathbb{E}(\varepsilon) = \mathbf{0}, \qquad \mathrm{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n.$$

Consequently,

$$\mathbb{E}(\mathbf{Y}) = \mathbf{X}\beta, \qquad \mathrm{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n.$$

Least Squares Does Not Require a Distributional Assumption

The least squares estimator is defined through the optimization problem

$$\min_{\beta} \|\mathbf{Y} - \mathbf{X}\beta\|_2^2 .$$

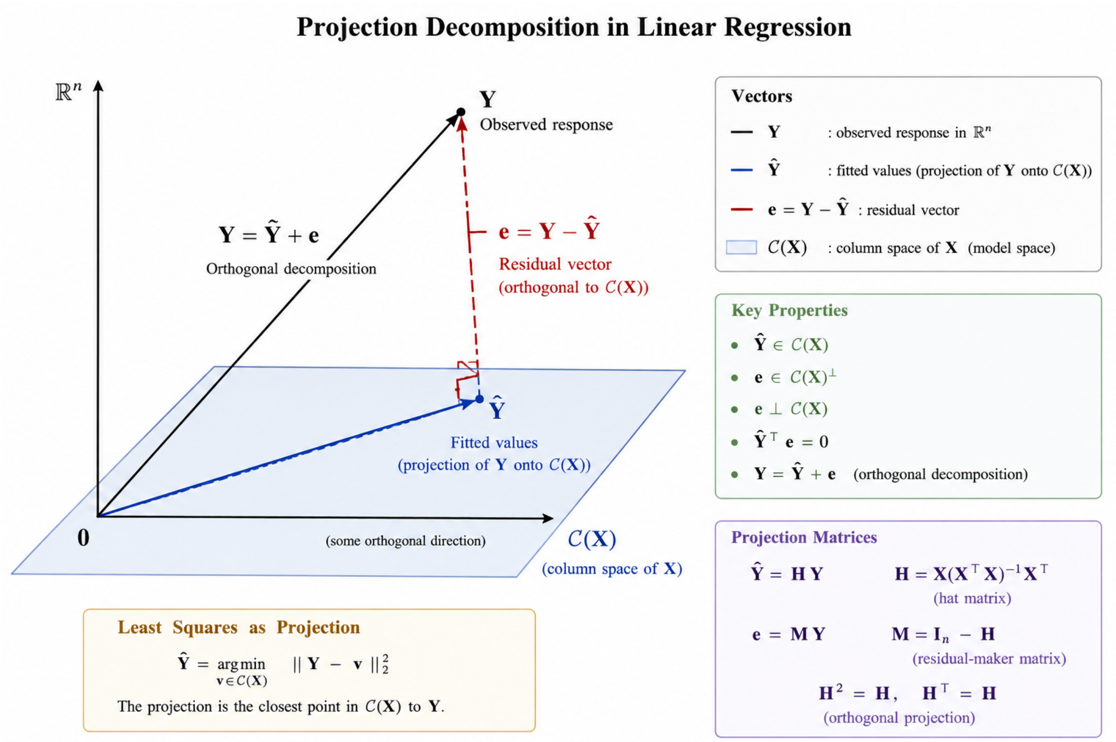


Figure 2.1: Geometric interpretation of least squares as an orthogonal projection.

No distributional assumption on ε is required to define or compute the least squares estimator.
Assumptions such as

$$\mathbb{E}(\varepsilon) = \mathbf{0}$$

and

$$\operatorname{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n$$

are needed when we study the **statistical properties** of the estimator.
Normality will be introduced later when we study exact finite-sample inference.

2.1 The Least Squares Criterion

Motivationß

For any candidate value β, the corresponding fitted vector is

$$\mathbf{X}\beta.$$

The discrepancy between the observed response $\mathbf{Y}$ and this candidate fit is

$$\mathbf{Y} - \mathbf{X}\beta.$$

A natural idea is to choose β so that this discrepancy is as small as possible.

To measure its size, we use the squared Euclidean norm.

Residual Sum of Squares

The **residual sum of squares** associated with a candidate β is

$$S(\beta) = \|\mathbf{Y} - \mathbf{X}\beta\|_2^2 .$$

Equivalently,

$$S(\beta) = (\mathbf{Y} - \mathbf{X}\beta)^\top (\mathbf{Y} - \mathbf{X}\beta).$$

In scalar notation,

$$S(\beta) = \sum_{i=1}^{n} \left(Y_i - \mathbf{x}_i^\top \beta\right)^2.$$

This last expression explains the name **least squares**: we choose β to make the sum of the squared discrepancies as small as possible.

We therefore define

$$\hat{\beta} = \arg\min_{\beta \in \mathbb{R}^p} S(\beta).$$

A **least squares estimator** is any value $\hat{\beta}$ satisfying

$$S(\hat{\beta}) \leq S(\beta)$$

for every $\beta \in \mathbb{R}^p$.

2.2 Derivation of the Least Squares Estimator

Expansion of the Criterion

Starting from

$$S(\beta) = (\mathbf{Y} - \mathbf{X}\beta)^\top (\mathbf{Y} - \mathbf{X}\beta),$$

we expand:

$$S(\beta) = \mathbf{Y}^\top \mathbf{Y} - \mathbf{Y}^\top \mathbf{X}\beta - \beta^\top \mathbf{X}^\top \mathbf{Y} + \beta^\top \mathbf{X}^\top \mathbf{X}\beta.$$

Because

$$\mathbf{Y}^\top \mathbf{X}\beta$$

is a scalar,

$$\mathbf{Y}^\top \mathbf{X}\beta = \beta^\top \mathbf{X}^\top \mathbf{Y}.$$

Therefore,

$$\boxed{S(\beta) = \mathbf{Y}^\top \mathbf{Y} - 2\beta^\top \mathbf{X}^\top \mathbf{Y} + \beta^\top \mathbf{X}^\top \mathbf{X}\beta.}$$

Derivation of the Normal Equations

Differentiate $S(\beta)$ with respect to β:

$$\frac{\partial S(\beta)}{\partial \beta} = -2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X}\beta.$$

At a minimum, the gradient is zero, so

$$-2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X}\hat{\beta} = \mathbf{0}.$$

Thus,

$$\boxed{\mathbf{X}^\top \mathbf{X}\hat{\beta} = \mathbf{X}^\top \mathbf{Y}.}$$

These are called the **normal equations**.

> **i** Why Are They Called the Normal Equations?
>
> The normal equations can be written as
>
> $$\mathbf{X}^\top \left(\mathbf{Y} - \mathbf{X}\hat{\beta}\right) = \mathbf{0}.$$
>
> Later we will define
>
> $$\mathbf{e} = \mathbf{Y} - \mathbf{X}\hat{\beta}.$$
>
> Therefore,
>
> $$\mathbf{X}^\top \mathbf{e} = \mathbf{0}.$$
>
> This means that $\mathbf{e}$ is perpendicular, or **normal**, to the column space of $\mathbf{X}$.
> Thus, the word *normal* is being used in its geometric sense.

Why is the Stationary Point a Minimum?

The Hessian of the least squares criterion is

$$\nabla^2 S(\beta) = 2\mathbf{X}^\top \mathbf{X}.$$

For any $\mathbf{a} \in \mathbb{R}^p$,

$$\mathbf{a}^\top \mathbf{X}^\top \mathbf{X}\mathbf{a} = (\mathbf{X}\mathbf{a})^\top (\mathbf{X}\mathbf{a}) = \|\mathbf{X}\mathbf{a}\|_2^2 \geq 0.$$

Therefore, $\mathbf{X}^\top \mathbf{X}$ is positive semidefinite, and $S(\beta)$ is a convex function.

If $\mathbf{X}$ has full column rank, then for every $\mathbf{a} \neq \mathbf{0}$,

$$\mathbf{X}\mathbf{a} \neq \mathbf{0},$$

and hence

$$\mathbf{a}^\top \mathbf{X}^\top \mathbf{X}\mathbf{a} > 0.$$

Thus, $\mathbf{X}^\top \mathbf{X}$ is positive definite and the least squares minimizer is unique.

A vector $\hat{\beta}$ minimizes

$$S(\beta) = \|\mathbf{Y} - \mathbf{X}\beta\|_2^2$$

if and only if it satisfies the normal equations

$$\mathbf{X}^\top \mathbf{X}\hat{\beta} = \mathbf{X}^\top \mathbf{Y}.$$

If $\mathbf{X}$ has full column rank,

$$\operatorname{rank}(\mathbf{X}) = p,$$

then $\mathbf{X}^\top \mathbf{X}$ is invertible and the least squares estimator is unique:

$$\boxed{\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.}$$

This is commonly called the **ordinary least squares estimator**, or **OLS estimator**.

When Is the Closed-Form Formula Valid?

The expression

$$(\mathbf{X}^\top \mathbf{X})^{-1}$$

exists if and only if $\mathbf{X}$ has full column rank:

$$\operatorname{rank}(\mathbf{X}) = p.$$

This means that the columns of $\mathbf{X}$ are linearly independent.
If they are linearly dependent, then $\mathbf{X}^\top\mathbf{X}$ is singular and the inverse does not exist.
We will return to this issue later.

2.3 Geometric Interpretation of Least Squares

The algebraic derivation gives us the estimator. However, the geometry explains **why** least squares has its important properties.

Recap: Column Space of the Design Matrix

The column space of $\mathbf{X}$ is

$$\mathcal{C}(\mathbf{X}) = \{\mathbf{X}\beta : \beta \in \mathbb{R}^p\}.$$

Thus, $\mathcal{C}(\mathbf{X})$ is the set of all response vectors that can be represented exactly by the linear model.
If $\mathbf{X}$ has full column rank p, then

$$\dim\{\mathcal{C}(\mathbf{X})\} = p.$$

Because $\mathbf{X}$ has n rows,

$$\mathcal{C}(\mathbf{X}) \subseteq \mathbb{R}^n.$$

2.3.1 Least Squares as Projection

The fitted value vector is

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta}.$$

Because

$$\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X}),$$

least squares chooses the vector in $\mathcal{C}(\mathbf{X})$ that is closest to the observed vector $\mathbf{Y}$.

That is,

$$\hat{\mathbf{Y}} = \arg \min_{\mathbf{v} \in \mathcal{C}(\mathbf{X})} \|\mathbf{Y} - \mathbf{v}\|_2^2.$$

Therefore, $\hat{\mathbf{Y}}$ is the **orthogonal projection** of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$.

After obtaining the least squares estimator, define the **residual vector**

$$\boxed{\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}} = \mathbf{Y} - \mathbf{X}\hat{\beta}.}$$

The individual residuals are

$$e_i = Y_i - \hat{Y}_i, \qquad i = 1, \dots, n.$$

Geometrically, $\mathbf{e}$ is the component of $\mathbf{Y}$ that is orthogonal to the model space $\mathcal{C}(\mathbf{X})$.

2.4 Orthogonality of the Residuals

Starting from the normal equations,

$$\mathbf{X}^\top \mathbf{X}\hat{\beta} = \mathbf{X}^\top \mathbf{Y},$$

we obtain

$$\mathbf{X}^\top \left(\mathbf{Y} - \mathbf{X}\hat{\beta}\right) = \mathbf{0}.$$

Since

$$\mathbf{e} = \mathbf{Y} - \mathbf{X}\hat{\beta},$$

we have

$$\boxed{\mathbf{X}^\top \mathbf{e} = \mathbf{0}.}$$

Suppose

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}^{(1)} & \mathbf{x}^{(2)} & \cdots & \mathbf{x}^{(p)} \end{pmatrix},$$

where $\mathbf{x}^{(j)}$ denotes column j of $\mathbf{X}$.

Then

$$\mathbf{X}^\top \mathbf{e} = \begin{pmatrix} (\mathbf{x}^{(1)})^\top \mathbf{e} \\ (\mathbf{x}^{(2)})^\top \mathbf{e} \\ \vdots \\ (\mathbf{x}^{(p)})^\top \mathbf{e} \end{pmatrix} = \mathbf{0}.$$

Therefore,

$$(\mathbf{x}^{(j)})^\top \mathbf{e} = 0, \qquad j = 1, \dots, p.$$

Hence, the residual vector is orthogonal to every column of $\mathbf{X}$ and therefore to every vector in $\mathcal{C}(\mathbf{X})$:

$$\boxed{\mathbf{e} \perp \mathcal{C}(\mathbf{X}).}$$

Least squares decomposes the response vector into

$$\boxed{\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}.}$$

Here,

$$\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$$

and

$$\mathbf{e} \in \mathcal{C}(\mathbf{X})^\perp.$$

Therefore,

$$\boxed{\hat{\mathbf{Y}}^\top \mathbf{e} = 0.}$$

This orthogonal decomposition is the geometric foundation of least squares regression.

2.4.1 Consequence when an Intercept Is Included

If the model includes an intercept, then one column of $\mathbf{X}$ is

$$\mathbf{1}_n = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}.$$

Since the residual vector is orthogonal to every column of $\mathbf{X}$,

$$\mathbf{1}_n^\top \mathbf{e} = 0.$$

Therefore,

$$\boxed{\sum_{i=1}^{n} e_i = 0.}$$

Since

$$e_i = Y_i - \hat{Y}_i,$$

we also have

$$\sum_{i=1}^{n} Y_i = \sum_{i=1}^{n} \hat{Y}_i.$$

Thus,

$$\boxed{\bar{\hat{Y}} = \bar{Y}.}$$

2.5 The Hat Matrix

In the full-column-rank case,

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta}.$$

Substituting

$$\hat{\beta} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y},$$

we obtain

$$\hat{\mathbf{Y}} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}.$$

Define

$$\boxed{\mathbf{H} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top.}$$

Then

$$\boxed{\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}.}$$

The matrix $\mathbf{H}$ is called the **hat matrix** because it puts the "hat" on $\mathbf{Y}$:

$$\mathbf{Y} \quad \longrightarrow \quad \mathbf{H}\mathbf{Y} = \hat{\mathbf{Y}}.$$

2.5.1 Properties of the Hat Matrix

When $\mathbf{X}$ has full column rank, the hat matrix satisfies:

i) Symmetry

$$\boxed{\mathbf{H}^\top = \mathbf{H}.}$$

ii) Idempotence

$$\boxed{\mathbf{H}^2 = \mathbf{H}.}$$

iii) Projection of the design matrix

$$\boxed{\mathbf{H}\mathbf{X} = \mathbf{X}.}$$

iv) Rank

$$\boxed{\operatorname{rank}(\mathbf{H}) = p.}$$

v) Trace

$$\boxed{\operatorname{tr}(\mathbf{H}) = p.}$$

A symmetric and idempotent matrix represents an orthogonal projection.

Therefore, $\mathbf{H}$ is the orthogonal projection matrix onto

$$\mathcal{C}(\mathbf{X}).$$

> **Why it is symmetric**
>
> We have
>
> $$\begin{aligned}\mathbf{H}^\top &= \left[\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\right]^\top \\ &= \mathbf{X}\left[(\mathbf{X}^\top\mathbf{X})^{-1}\right]^\top \mathbf{X}^\top.\end{aligned}$$
>
> Since $\mathbf{X}^\top\mathbf{X}$ is symmetric, its inverse is also symmetric. Therefore,
>
> $$\mathbf{H}^\top = \mathbf{H}.$$

> **Why it is idempotent**
>
> We have
>
> $$\begin{aligned}\mathbf{H}^2 &= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top \\ &= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top \\ &= \mathbf{H}.\end{aligned}$$
>
> Thus,
>
> $$\mathbf{H}^2 = \mathbf{H}.$$

What Does Idempotence Mean Geometrically?

If we project $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$ once, we obtain

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}.$$

Projecting the fitted vector again does nothing:

$$\mathbf{H}\hat{\mathbf{Y}} = \mathbf{H}^2\mathbf{Y} = \mathbf{H}\mathbf{Y} = \hat{\mathbf{Y}}.$$

Once a vector is already inside the model space, projecting it onto the same space leaves it unchanged.

3 The Residual-Maker Matrix

Since

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$$

and

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y},$$

we have

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\mathbf{Y}.$$

Define

$$\boxed{\mathbf{M} = \mathbf{I}_n - \mathbf{H}.}$$

Then

$$\boxed{\mathbf{e} = \mathbf{M}\mathbf{Y}.}$$

The matrix $\mathbf{M}$ is called the **residual-maker matrix**.

It removes the component of $\mathbf{Y}$ lying in $\mathcal{C}(\mathbf{X})$.

3.1 Properties of the Residual-Maker Matrix

Since

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H},$$

we obtain

$$\mathbf{M}^\top = \mathbf{M}$$

and

$$\mathbf{M}^2 = \mathbf{M}.$$

Thus, $\mathbf{M}$ is also an orthogonal projection matrix.

While $\mathbf{H}$ projects onto

$$\mathcal{C}(\mathbf{X}),$$

$\mathbf{M}$ projects onto

$$\mathcal{C}(\mathbf{X})^\perp.$$

In addition,

$$\boxed{\mathbf{HM} = \mathbf{MH} = \mathbf{0}.}$$

Under full column rank,

$$\operatorname{rank}(\mathbf{M}) = n - p$$

and

$$\operatorname{tr}(\mathbf{M}) = n - p.$$

Two Complementary Projection Matrices

The matrices $\mathbf{H}$ and $\mathbf{M}$ split $\mathbb{R}^n$ into two orthogonal components:

$$\mathbb{R}^n = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}(\mathbf{X})^{\perp}.$$

For the observed response,

$$\mathbf{Y} = \underbrace{\mathbf{HY}}_{\hat{\mathbf{Y}}} + \underbrace{\mathbf{MY}}_{\mathbf{e}}.$$

Thus,

$$\boxed{\mathbf{H} + \mathbf{M} = \mathbf{I}_n.}$$

4 Sum of Squares Decomposition

4.1 Uncentered Geometric Decomposition

Since

$$\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}$$

and

$$\hat{\mathbf{Y}}^\top \mathbf{e} = 0,$$

we have

$$\begin{aligned}
\mathbf{Y}^\top \mathbf{Y} &= (\hat{\mathbf{Y}} + \mathbf{e})^\top (\hat{\mathbf{Y}} + \mathbf{e}) \\
&= \hat{\mathbf{Y}}^\top \hat{\mathbf{Y}} + 2\hat{\mathbf{Y}}^\top \mathbf{e} + \mathbf{e}^\top \mathbf{e} \\
&= \hat{\mathbf{Y}}^\top \hat{\mathbf{Y}} + \mathbf{e}^\top \mathbf{e}.
\end{aligned}$$

Therefore,

$$\boxed{\|\mathbf{Y}\|_2^2 = \|\hat{\mathbf{Y}}\|_2^2 + \|\mathbf{e}\|_2^2.}$$

This is a direct application of the Pythagorean theorem.

4.2 Centered Sum of Squares Decomposition

When the regression model contains an intercept,

$$\bar{\hat{Y}} = \bar{Y}.$$

Therefore,

$$Y_i - \bar{Y} = (\hat{Y}_i - \bar{Y}) + (Y_i - \hat{Y}_i).$$

The two components are orthogonal, giving

$$\boxed{\sum_{i=1}^{n}(Y_i - \bar{Y})^2 = \sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2.}$$

These quantities will later be called

- **total sum of squares**
$$\mathrm{SST} = \sum_{i=1}^{n}(Y_i - \bar{Y})^2;$$
- **regression sum of squares**
$$\mathrm{SSR} = \sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2;$$
- **error sum of squares**
$$\mathrm{SSE} = \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2.$$

Thus,

$$\boxed{\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}.}$$

! Centered vs. Uncentered Decomposition

The geometric identity

$$\mathbf{Y}^\top \mathbf{Y} = \hat{\mathbf{Y}}^\top \hat{\mathbf{Y}} + \mathbf{e}^\top \mathbf{e}$$

is an **uncentered** decomposition.

The familiar regression decomposition

$$\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$$

is a **centered** decomposition and relies on the model containing an intercept. These two decompositions should not be confused.

4.3 Example: Simple Linear Regression

Consider the simple regression dataset

$$\begin{array}{c|cccc} x_i & 0 & 1 & 2 & 3 \\ \hline y_i & 1 & 3 & 3 & 5 \end{array}.$$

We fit the model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

The response vector is

$$\mathbf{Y} = \begin{pmatrix} 1 \\ 3 \\ 3 \\ 5 \end{pmatrix},$$

and the design matrix is

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}.$$

Step 1: Compute $\mathbf{X}^\top \mathbf{X}$

We have

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix}.$$

Step 2: Compute $\mathbf{X}^\top \mathbf{Y}$

We obtain

$$\mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} 12 \\ 24 \end{pmatrix}.$$

Step 3: Compute $\widehat{\beta}$

Since

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{20} \begin{pmatrix} 14 & -6 \\ -6 & 4 \end{pmatrix},$$

the least squares estimator is

$$\begin{aligned} \widehat{\beta} &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y} \\ &= \frac{1}{20} \begin{pmatrix} 14 & -6 \\ -6 & 4 \end{pmatrix} \begin{pmatrix} 12 \\ 24 \end{pmatrix} \\ &= \begin{pmatrix} 1.2 \\ 1.2 \end{pmatrix}. \end{aligned}$$

Thus,

$$\widehat{\beta}_0 = 1.2, \qquad \widehat{\beta}_1 = 1.2,$$

and the fitted regression line is

$$\boxed{\hat{Y} = 1.2 + 1.2x.}$$

Step 4: Compute the Fitted Values

The fitted values are

$$\hat{\mathbf{Y}} = \mathbf{X}\widehat{\beta} = \begin{pmatrix} 1.2 \\ 2.4 \\ 3.6 \\ 4.8 \end{pmatrix}.$$

Step 5: Compute the Residuals

The residual vector is

$$
\begin{aligned}
\mathbf{e} &= \mathbf{Y} - \hat{\mathbf{Y}} \\
&= \begin{pmatrix} 1 \\ 3 \\ 3 \\ 5 \end{pmatrix} - \begin{pmatrix} 1.2 \\ 2.4 \\ 3.6 \\ 4.8 \end{pmatrix} \\
&= \begin{pmatrix} -0.2 \\ 0.6 \\ -0.6 \\ 0.2 \end{pmatrix}.
\end{aligned}
$$

Notice that

$$\sum_{i=1}^{4} e_i = -0.2 + 0.6 - 0.6 + 0.2 = 0.$$

This occurs because the model contains an intercept.

Step 6: Verify Orthogonality

We compute

$$\mathbf{X}^\top \mathbf{e} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} -0.2 \\ 0.6 \\ -0.6 \\ 0.2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Thus,

$$\boxed{\mathbf{X}^\top \mathbf{e} = \mathbf{0}.}$$

We can also verify that

$\hat{\mathbf{Y}}^\top \mathbf{e} = 0.$

Step 7: Compute the Hat Matrix

The hat matrix is

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top.$$

For this example,

$$\mathbf{H} = \begin{pmatrix} 0.7 & 0.4 & 0.1 & -0.2 \\ 0.4 & 0.3 & 0.2 & 0.1 \\ 0.1 & 0.2 & 0.3 & 0.4 \\ -0.2 & 0.1 & 0.4 & 0.7 \end{pmatrix}.$$

Multiplying by $\mathbf{Y}$ gives

$$\mathbf{HY} = \begin{pmatrix} 1.2 \\ 2.4 \\ 3.6 \\ 4.8 \end{pmatrix} = \hat{\mathbf{Y}}.$$

Thus, the hat matrix projects the observed response onto the model space.

We can verify the previous calculations in R.

```r
x <- c(0, 1, 2, 3)
y <- c(1, 3, 3, 5)

df <- data.frame(
  x = x,
  y = y
)

fit <- lm(y ~ x, data = df)

coef(fit)
```

```
(Intercept)           x
        1.2         1.2
```

The estimated coefficients are

$$\hat{\beta}_0 = 1.2, \qquad \hat{\beta}_1 = 1.2.$$

We can also compute the estimator directly using matrix operations.

```r
X <- model.matrix(fit)
X
```

```
  (Intercept) x
1           1 0
2           1 1
3           1 2
4           1 3
attr(,"assign")
[1] 0 1
```

```
Y <- matrix(y, ncol = 1)

beta_hat <-
  solve(t(X) %*% X) %*%
  t(X) %*%
  Y

beta_hat
```

```
            [,1]
(Intercept)  1.2
x            1.2
```

Compute the fitted values and residuals:

```
y_hat <- X %*% beta_hat
e <- Y - y_hat

y_hat
```

```
  [,1]
1  1.2
2  2.4
3  3.6
4  4.8
```

```
e
```

```
  [,1]
1 -0.2
2  0.6
3 -0.6
4  0.2
```

Check that the residuals sum to zero:

```r
sum(e)
```

```
[1] -3.774758e-15
```

Compute the hat matrix:

```r
H <-
  X %*%
  solve(t(X) %*% X) %*%
  t(X)

round(H, 4)
```

```
     1   2   3    4
1  0.7 0.4 0.1 -0.2
2  0.4 0.3 0.2  0.1
3  0.1 0.2 0.3  0.4
4 -0.2 0.1 0.4  0.7
```

Verify symmetry:

```r
round(H - t(H), 10)
```

```
  1 2 3 4
1 0 0 0 0
2 0 0 0 0
3 0 0 0 0
4 0 0 0 0
```

Verify idempotence:

```r
round(H %*% H - H, 10)
```

```
  1 2 3 4
1 0 0 0 0
2 0 0 0 0
3 0 0 0 0
4 0 0 0 0
```

Verify the normal equations geometrically:

```
round(t(X) %*% e, 10)
```

```
            [,1]
(Intercept)    0
x              0
```

Verify that fitted values and residuals are orthogonal:

```
round(t(y_hat) %*% e, 10)
```

```
     [,1]
[1,]    0
```

Construct the residual-maker matrix:

```
M <- diag(nrow(X)) - H

round(M, 4)
```

```
     1    2    3    4
1  0.3 -0.4 -0.1  0.2
2 -0.4  0.7 -0.2 -0.1
3 -0.1 -0.2  0.7 -0.4
4  0.2 -0.1 -0.4  0.3
```

```
round(M %*% M - M, 10)
```

```
  1 2 3 4
1 0 0 0 0
2 0 0 0 0
3 0 0 0 0
4 0 0 0 0
```

Verify that

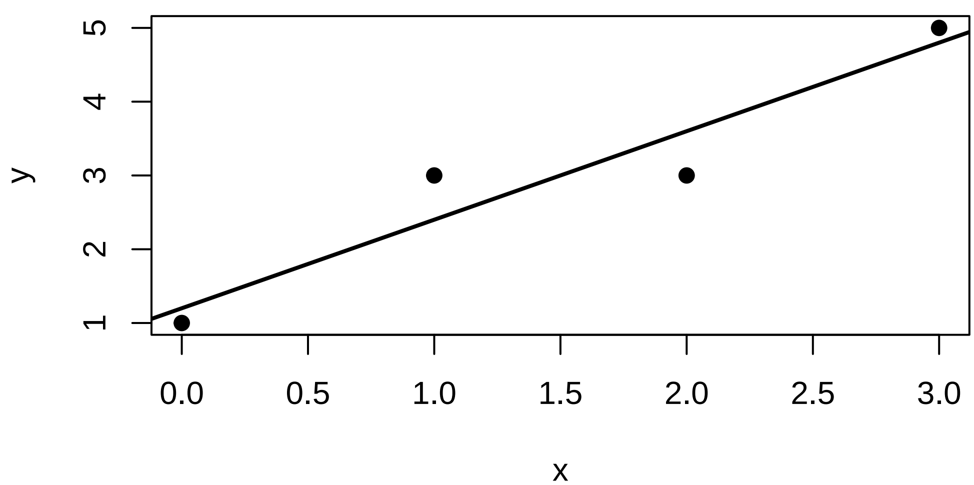
$$\mathbf{e} = \mathbf{MY}.$$

```r
round(M %*% Y - e, 10)
```

```
  [,1]
1    0
2    0
3    0
4    0
```

Finally, plot the fitted regression line:

```r
plot(
  x,
  y,
  pch = 19,
  xlab = "x",
  ylab = "y"
)

abline(
  fit,
  lwd = 2
)
```



5 What Happens Without Full Column Rank?

Suppose

$$\operatorname{rank}(\mathbf{X}) < p.$$

Then the columns of $\mathbf{X}$ are linearly dependent, and

$$\mathbf{X}^\top \mathbf{X}$$

is singular.

Therefore,

$$(\mathbf{X}^\top \mathbf{X})^{-1}$$

does not exist.

! A Subtle but Important Point

When $\mathbf{X}$ does not have full column rank,

- a least squares solution still exists;
- $\hat{\beta}$ may not be unique;
- the fitted vector $\hat{\mathbf{Y}}$ is still unique;
- the residual vector $\mathbf{e}$ is still unique.

Thus,

$$\boxed{\hat{\beta} \text{ may not be unique, but } \hat{\mathbf{Y}} \text{ is unique.}}$$

The reason is geometric: even if several coefficient vectors represent the same point in $\mathcal{C}(\mathbf{X})$, the orthogonal projection of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$ is unique.
We will discuss rank deficiency, generalized inverses, and estimability in more detail later.

6 Interpretation of the Geometry

The observed vector

$$\mathbf{Y} \in \mathbb{R}^n.$$

When $\mathbf{X}$ has full column rank p, the model space

$$\mathcal{C}(\mathbf{X})$$

is a p-dimensional subspace of $\mathbb{R}^n$.

Least squares finds the point in this subspace that is closest to $\mathbf{Y}$.

The central relationships are

$$\boxed{\min_{\beta} \|\mathbf{Y} - \mathbf{X}\beta\|_2^2}$$

$$\Downarrow$$

$$\boxed{\mathbf{X}^\top(\mathbf{Y} - \mathbf{X}\widehat{\beta}) = \mathbf{0}}$$

$$\Downarrow$$

$$\boxed{\mathbf{e} \perp \mathcal{C}(\mathbf{X})}$$

$$\Downarrow$$

$$\boxed{\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}}$$

$$\Downarrow$$

$$\boxed{\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}.}$$

Thus,

- fitted values are orthogonal projections;
- residuals are orthogonal to the model space;
- the hat matrix is a projection matrix;
- the residual-maker matrix projects onto the orthogonal complement;
- sum-of-squares decompositions are consequences of the Pythagorean theorem.

7 Looking Ahead: Statistical Properties of OLS

Looking Ahead

Everything we have done so far is primarily **algebraic and geometric**.
We defined

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$$

by solving an optimization problem.
However, under the statistical model,

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon,$$

the response $\mathbf{Y}$ is random.
Therefore,

$$\hat{\beta}$$

is also a random vector.
In next chapter, we will study quantities such as

$$\mathbb{E}(\hat{\beta})$$

and

$$\mathrm{Var}(\hat{\beta}),$$

followed by estimation of σ^2, sampling distributions, confidence intervals, and hypothesis testing.

8 In-Class Discussion Questions

1. Why does minimizing

$$\|\mathbf{Y} - \mathbf{X}\beta\|_2^2$$

   lead to an orthogonality condition?

2. What is the geometric meaning of

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0}?$$

3. Why are the normal equations called **normal** equations?

4. Why is the hat matrix called a projection matrix?

5. Why does

$$\mathbf{H}^2 = \mathbf{H}$$

   make sense geometrically?

6. Why does including an intercept imply

$$\sum_{i=1}^{n} e_i = 0?$$

7. What is the difference between the roles of

$$\mathbf{H}$$

   and

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}?$$

8. If $\mathbf{X}$ does not have full column rank, why can the fitted values still be unique even when $\widehat{\beta}$ is not?

9 Practice Problems

9.1 Conceptual Problems

1. Explain in words what the column space

$$\mathcal{C}(\mathbf{X})$$

represents in a linear regression model.

2. Explain why

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0}$$

is both an algebraic and a geometric statement.

3. Explain why

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$$

is an orthogonal projection.

4. Explain the difference between

$$\mathbf{H}$$

and

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}.$$

5. Why does the least squares estimator not require normal errors?

6. Why is full column rank needed for

$$(\mathbf{X}^\top \mathbf{X})^{-1}$$

to exist?

7. Explain why

$$\hat{\beta}$$

may fail to be unique while

$$\hat{\mathbf{Y}}$$

remains unique.

9.2 Computational Problems

Let

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \mathbf{Y} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}.$$

1. Compute $\mathbf{X}^\top \mathbf{X}$.
2. Compute $\mathbf{X}^\top \mathbf{Y}$.
3. Find $\hat{\beta}$.
4. Compute $\hat{\mathbf{Y}}$.
5. Compute the residual vector $\mathbf{e}$.
6. Verify that

   $\mathbf{X}^\top \mathbf{e} = \mathbf{0}$.
7. Compute the hat matrix $\mathbf{H}$.
8. Verify that $\mathbf{H}^\top = \mathbf{H}$ and $\mathbf{H}^2 = \mathbf{H}$.
9. Compute $\mathbf{M} = \mathbf{I}_n - \mathbf{H}$.
10. Verify that $\mathbf{e} = \mathbf{MY}$.

9.3 Proof-Based Problems

1. Prove that

$$\mathbf{X}^\top \mathbf{X}$$

is positive semidefinite.

2. Show that if $\mathbf{X}$ has full column rank, then

$$\mathbf{X}^\top \mathbf{X}$$

is positive definite.

3. Prove that the hat matrix

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top$$

is symmetric.

4. Prove that $\mathbf{H}$ is idempotent.

5. Prove that

$$\mathbf{H}\mathbf{X} = \mathbf{X}.$$

6. Prove that

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}$$

is symmetric and idempotent.

7. Show that

$$\mathbf{H}\mathbf{M} = \mathbf{0}.$$

10 Suggested Homework

Complete the following tasks:

1. Derive the normal equations from

$$S(\beta) = \|\mathbf{Y} - \mathbf{X}\beta\|_2^2.$$

2. Show that the least squares solution satisfies

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0}.$$

3. Explain why this implies that $\hat{\mathbf{Y}}$ is the orthogonal projection of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$.
4. Prove that the hat matrix is symmetric and idempotent.
5. Prove that the residual-maker matrix is symmetric and idempotent.
6. Fit a simple regression model in R and compute:
   - $\widehat{\beta}$;
   - $\hat{\mathbf{Y}}$;
   - $\mathbf{e}$;
   - $\mathbf{H}$;
   - $\mathbf{M}$.
7. Numerically verify in R that

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0},$$

$$\mathbf{H}^2 = \mathbf{H},$$

and

$$\mathbf{M}^2 = \mathbf{M}.$$

11 Summary

In this week, we introduced the least squares estimator through the optimization problem

$$\hat{\beta} = \arg\min_{\beta} \|\mathbf{Y} - \mathbf{X}\beta\|_2^2 .$$

Differentiating the residual sum of squares leads to the normal equations

$$\boxed{\mathbf{X}^\top \mathbf{X}\hat{\beta} = \mathbf{X}^\top \mathbf{Y}.}$$

When $\mathbf{X}$ has full column rank,

$$\boxed{\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \mathbf{Y}.}$$

The fitted response is

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y},$$

where

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top$$

is the hat matrix.

Geometrically,

$$\hat{\mathbf{Y}}$$

is the orthogonal projection of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$.

The residual vector

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$$

lies in the orthogonal complement of the model space, so

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0}.$$

Therefore,

$$\boxed{\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}}$$

is an orthogonal decomposition.

In next chapter, we will treat $\widehat{\beta}$ as a random vector and study its expectation, variance, sampling distribution, and the foundations of statistical inference for linear models.

Side Fact: Matrix Calculus Facts Used in this chapter

Let β be a column vector.

For a constant vector $\mathbf{a}$,

$$\frac{\partial}{\partial \beta} (\mathbf{a}^\top \beta) = \mathbf{a}.$$

For a constant matrix $\mathbf{A}$,

$$\frac{\partial}{\partial \beta} \left(\beta^\top \mathbf{A} \beta\right) = (\mathbf{A} + \mathbf{A}^\top)\beta.$$

If $\mathbf{A}$ is symmetric,

$$\mathbf{A}^\top = \mathbf{A},$$

and therefore

$$\boxed{\frac{\partial}{\partial \beta} \left(\beta^\top \mathbf{A} \beta\right) = 2\mathbf{A}\beta.}$$

These identities justify the derivative used in the least squares criterion.

References

Montgomery, Douglas C. 2017. *Design and Analysis of Experiments*. John wiley & Sons.

Montgomery, Douglas C, Elizabeth A Peck, and G Geoffrey Vining. 2021. *Introduction to Linear Regression Analysis*. John Wiley & Sons.

Petersen, Kaare Brandt, and Michael Syskind Pedersen. 2008. *The Matrix Cookbook*. Technical University of Denmark.

Seber, George AF, and Alan J Lee. 2003. *Linear Regression Analysis*. John Wiley & Sons.

Part I

Appendix

Matrix Algebra Review

Matrix dimensions

If $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times p$, then $\mathbf{AB}$ is defined and is $m \times p$.

Transpose rules

For conformable matrices,

$$(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top.$$

Inverse rule

If $\mathbf{A}$ and $\mathbf{B}$ are invertible, then

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}.$$

Symmetric matrices

A matrix $\mathbf{A}$ is symmetric if

$$\mathbf{A}^\top = \mathbf{A}.$$

Covariance matrices are always symmetric.

A good reference of the linear algebra can be found from Petersen and Pedersen (2008).

- Petersen, Kaare Brandt, and Michael Syskind Pedersen. (2008). *The Matrix Cookbook*. Technical University of Denmark.